

# Implementing a Tyrer-Cuzick-Based Breast Cancer Risk Stratification Workflow in a Resident Primary Care Clinic: A Single-Center Quality Improvement Initiative

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## Background

Early identification of women at increased risk for breast cancer remains an underutilized opportunity in primary care, particularly prior to initiation of routine screening mammography. Contemporary guidelines from the American College of Radiology (ACR) and American Cancer Society (ACS) recommend enhanced screening, including annual breast MRI, for women with a calculated lifetime breast cancer risk  $\geq 20\%$ , known hereditary cancer predisposition, or prior chest radiation at a young age. Additionally, breast density modifies both cancer risk and mammographic sensitivity. Despite these recommendations, structured risk assessment is not routinely incorporated into primary care workflows, resulting in missed opportunities to identify high-risk patients earlier.

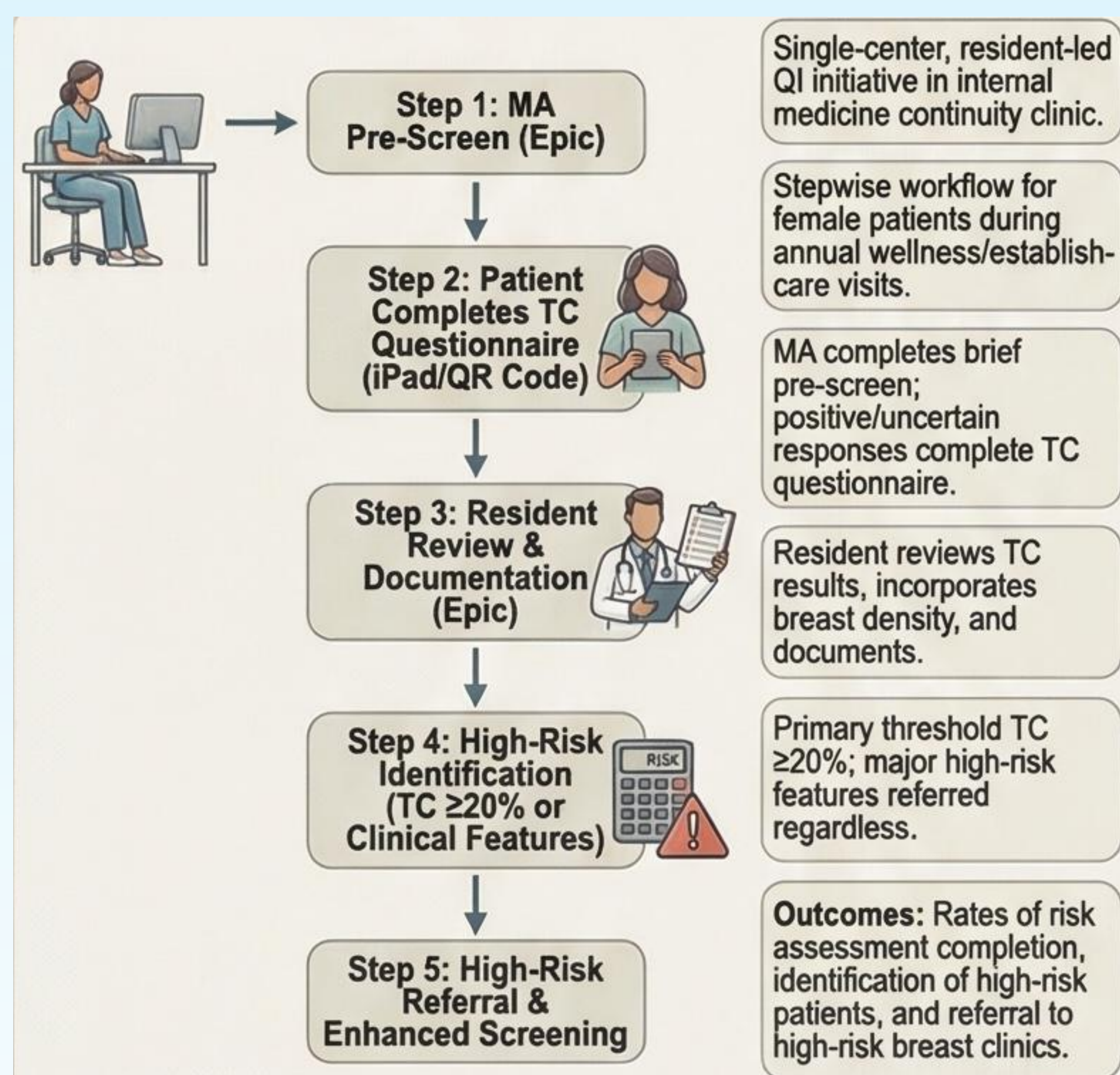
## Objectives

The primary objective is to develop and implement a standardized workflow integrating Tyrer-Cuzick (TC)-based breast cancer risk assessment into a resident primary care clinic. Secondary objectives include:

- Improving identification of patients eligible for enhanced breast cancer screening (e.g., annual MRI)
- Evaluating feasibility and workflow integration within a resident clinic setting
- Assessing utilization of high-risk referral pathways based on structured risk assessment

## Proposed Methods

This is a single-center, resident-led quality improvement initiative conducted in an internal medicine continuity clinic. A stepwise workflow will be implemented for female patients presenting for annual wellness or establish-care visits.



## Anticipated Contributions

This study aims to establish a feasible, scalable model for integrating breast cancer risk stratification into routine primary care. By embedding structured risk assessment into clinic workflows, this initiative has the potential to improve early identification of high-risk patients, increase appropriate utilization of enhanced screening, and align primary care practices with contemporary risk-based screening guidelines. It may also serve as a model for implementation in other residency programs and health systems.

